

Collision Avoidance for Continuum Robot Using FABRIK Algorithm

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Abstract—Continuum robots are manipulators able to bend at any point along their structure. This feature allows continuum robots to work in areas with complex geometry. In this regard, they are used as instruments and endoscopes in industry and medicine. Inverse kinematics for continuum robots is using for control and path planning. Recently, Forward And Backward Reaching Inverse Kinematics (FABRIK) algorithm has been used for solving inverse kinematics issue for continuum robots. For this, arcs of continuum robot are simplified to virtual links. But virtual links can not be used to define the distance between continuum robot body and obstacle without restoring whole robot section. In this paper, we propose a method for obstacle avoidance for multi-section continuum robot using FABRIK algorithm. The simulation results for the planar case show that the proposed method is able to avoid unwanted collisions.

Keywords—continuum robot, FABRIK, kinematics, collision avoidance

I. INTRODUCTION

Continuum robots are manipulators with ϕ continuously bending structure. Compared with traditional discrete rigid robots that consist of a finite number of rigid links connected together by joints, continuum robots do not contain rigid links (their length tends to zero) and has an infinite number of rotational joints [1]. These features make continuum robots able to bend at any point along their structure, hence, gives them the ability to avoid an unwanted collision in a complicated working area or grasp objects using their body. In industry, these robots are applied to non-destructive testing and repair gas turbine engines [2], nuclear reactors [3], for inspection in space [4] and as manipulators for laser cutting [5]. In medicine, they are widely used as endoscopes and surgical instruments for minimally invasive surgery [6].

Recently, Forward and Backward Reaching Inverse Kinematics (FABRIK) algorithm was applied to multi-section continuum robots [7,8]. Originally FABRIK was developed to work only with rigid-links robots, so FABRIK based algorithms simplify each section arc to one (arc chord) or two (arc tangents) virtual rigid link, so FABRIK is able to solve the inverse kinematics issue. Simulations show that proposed algorithms are faster than Jacobian based approach, finds a solution in most cases within the workspace and find a solution

when Jacobian based approach fails. However, obstacle avoidance methods for algorithms mentioned above have not been formulated.

Since continuum robots are designed to work in a confined space, the ability of their kinematics to avoid unwanted collisions is critical for motion planning. Virtual links used in FABRIK based solutions have only several points coinciding with the robot – the beginning and the end of each section. It means that it is not clear how far the robot body from obstacles until the curve is restored. If bending angle θ is small, virtual links can be used as approximation or envelope of robot single section (Figure 1a). This allows the use of virtual links to estimate the distance between the robots and the obstacle. But with an increase in the bending angle, the error of such a method will increase (Figure 1b). After bending more than 180° , this method will stop being reliable since the section will be outside the envelope (Figure 1c). This limits the algorithm to work only in the areas without obstacles. This paper presents a method for collision avoidance for multi-section continuum robot based on FABRIK algorithm.

II. KINEMATIC MODEL

The kinematic model for a single section of continuum robot is based on a constant curvature approach. This is a

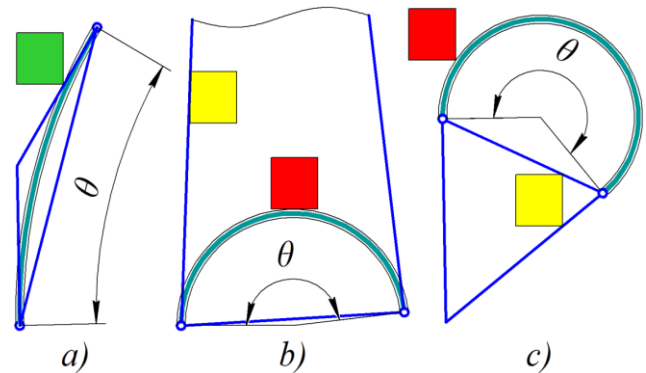


Fig. 1. Collisions of the continuum robot (green curve) and virtual links (blue lines) with obstacles (squares). a) $\theta \ll 180^\circ$. b) $\theta < 180^\circ$. c) $\theta > 180^\circ$. Green square – collision with virtual link close to the robot body. Yellow square – collision with virtual link far from the robot body. Red square – collision with robot body far from the virtual links.

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simple method that sufficiently accurately describes a continuum robot. Transformation matrix T_{TIP} that describe the frame of section end-effector are defined as follows:

$$T_{TIP} = T_{Base} T_{Rot}(\varphi) T_{Bend}\left(\frac{\theta}{2}\right) T_{Tr}(H) T_{Bend}\left(\frac{\theta}{2}\right) \quad (1)$$

where T_{Base} is section base frame, T_{Rot} is a rotation around the Z axis, φ is angle that define the spatial orientation of the curve plane, T_{Bend} is a rotation around the Y axis, θ is the bending angle, T_{Tr} is a translation along the Z axis and H is the chord length that defined as follows:

$$H = 2 \cdot \sin(\theta/2) \cdot S/\theta \quad (2)$$

where S is section length.

Multi-section kinematic models created by chaining together the models for the individual sections:

$$T_{TIP} = T_{Base} \prod_{i=1}^Q \left[T_{Rot}(\varphi_i) T_{Bend}\left(\frac{\theta_i}{2}\right) T_{Tr}(H_i) T_{Bend}\left(\frac{\theta_i}{2}\right) \right] \quad (3)$$

where Q is a number of robot sections.

Inverse kinematics of single section is also defined by constant curvature approach. Rotation angle $\varphi = 0$ for planar case and bending angle θ and is defined as:

$$\theta = 2 \arccos\left(\frac{Z}{\sqrt{X^2 + Y^2 + Z^2}}\right) \quad (4)$$

where X, Y and Z is end-effector coordinates in section base frame.

III. MULTI-SECTION INVERSE KINEMATICS

To describe the inverse kinematics algorithm for multi-section continuum robot was used the approach presented in

[7]. The FABRIK algorithm searches for a set of the bending angles θ and the rotation angles φ such that end-effector frame T_{TIP} fits with desired target-frame T_t (Figure 2). To implement FABRIK to continuum robot each section arc is replaced by virtual link – chords that connect the beginning and the end of the arc. Since chords can have any mutual spatial arrangement, we can assume that they connected by spherical joints. Chord length depends on the bending angle.

The algorithm finds a solution in several iterations. Each iteration includes a forward reaching, backward reaching and, if required, tip orientation adjustment. At the end of backward reaching it is checked if the end-effector frame reaches the target-frame: distance between the end-effector position and the target-frame position (t in Figure 2) must be lower than the given linear tolerance, and angles between the end-effector axes and target-frame axes (X_t, Y_t, Z_t in Figure 2) must be lower than the angular tolerance. Note, if the angular tolerance is higher than 180° , it means that algorithm does not consider end-effector orientation. If both conditions are true, the algorithm has reached the target-frame with the given accuracy.

A. Forward Reaching

Calculation of the forward reaching is carried out as a chain of links with constant lengths connected together by joints (Figure 2b). This stage adjusts chords to make it fit with the target-frame position. Tip of the robot is located at the target-point $P_Q^F = t$ and the other end-points can be calculated as follows:

$$P_i^F = (1 - \lambda_i) \cdot P_{i+1}^F + \lambda_i \cdot P_i^B \quad (5)$$

$$\lambda_i = H_{i+1} / (P_{i+1}^F - P_i^B) \quad (6)$$

where $i = 0 \dots Q - 1$. Hereinafter, the superscript {B} denotes

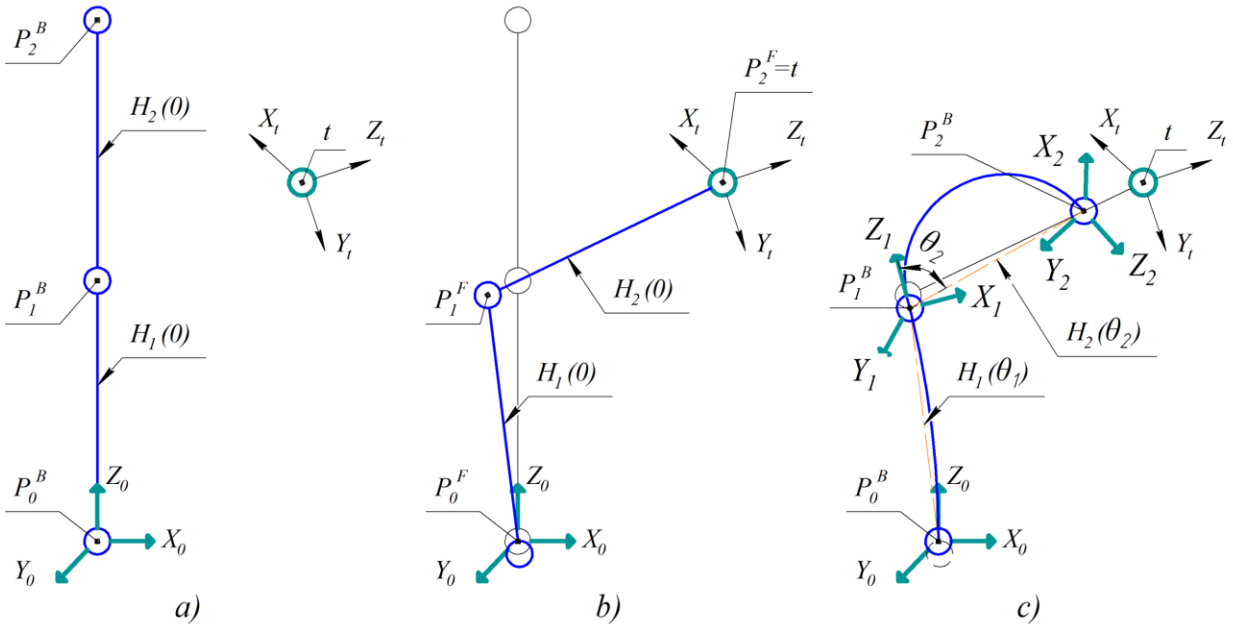


Fig. 2. Example of full iteration of the FABRIK algorithm for two-section continuum robot. a) the robot initial position and the target-frame (t – target-frame position, X, Y, Z – target-frame axis). b) the result of the forward reaching: tip placed at the target position and new positions of the endpoints defined. c) is the result of the backward reaching: curves are restored from chords, lengths of the chords renewed, tip frame defined.

the point obtained as a result of the backward reaching, and the superscript {F} is the points obtained as a result of the forward reaching. Bypass points are made from the end to the beginning.

B. Backward Reaching

The base-point is returned in its initial position $P_0^B = P_0$. The calculation of the rest points in backward reaching is carried out by equation (1) for bending angles θ defined using equation (4). Matrix T_{TIP} of current section becomes the base matrix T_{Base} for the next section. This provides a smooth transition between the bending sections. Lengths of the chords are renewed for using in the next iteration of forward reaching. The end-effector frame is defined when all robot curves are restored (Figure 2c). During this step, it is checked whether the distance between the end-effector position and the target-frame position lay inside tolerance. Bypass points are made from the beginning to the end.

IV. COLLISION DETECTION AND AVOIDANCE

Collision detection and avoidance occur during backward reaching after section T_{TIP} is defined. In this section we present collision detection and avoidance for planar case $\varphi = 0$. Constant curvature approach defines the continuum robot shape as an arc with constant length. So we use the arc center of each section to define the distance between obstacles and the continuum robot (Figure 3). At first we define arc center:

$$C = T_{MID} + S \cdot (G/|G|)/\theta \quad (7)$$

$$G = P_{i-1} + (P_i - P_{i-1})/2 - T_{MID} \quad (8)$$

T_{MID} is the position of arc midpoint defined as follows:

$$T_{MID} = T_{Base} T_{Bend} \left(\frac{\theta}{4} \right) T_{Tr}(H') \quad (9)$$

$$H' = 2 \cdot \sin(\theta/4) \cdot S/\theta \quad (10)$$

Then we define the safe zone. If an obstacle is in the safe zone, a collision has occurred. Do to that we define the equations of lines $a_i(x)$ and $a_{i-1}(x)$ that define the sector of the circle in which the arc lies:

$$k_i = \text{atan} \left(\frac{P_i^{(z)} - C^{(z)}}{P_i^{(x)} - C^{(x)}} \right) + \varepsilon \quad (11)$$

$$k_{i-1} = \text{atan} \left(\frac{P_{i-1}^{(z)} - C^{(z)}}{P_{i-1}^{(x)} - C^{(x)}} \right) - \varepsilon \quad (12)$$

$$a_i(x) = \tan(k_i) \cdot (x - C^{(x)}) + C^{(z)} \quad (13)$$

$$a_{i-1}(x) = \tan(k_{i-1}) \cdot (x - C^{(x)}) + C^{(z)} \quad (14)$$

where ε is a value that define length of the safe zone and depends on bending angle θ . Superscripts {x} and {z} denote x and z coordinates accordingly.

The next step is to remove obstacle points that do not meet the following condition:

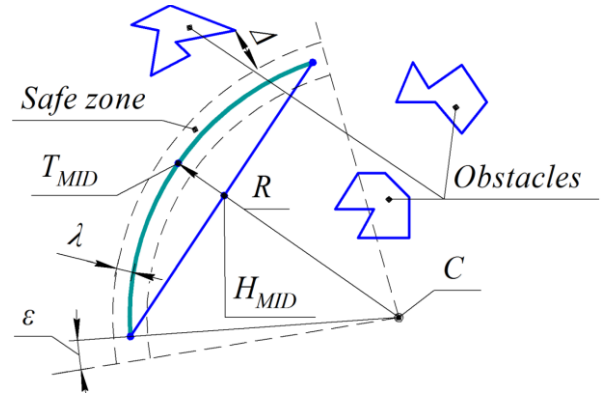


Fig. 3. Collision detection scheme.

$$\begin{cases} \text{sign}(O_j^{(z)} - a_i(O_j^{(x)})) = \text{sign}(T_{MID}^{(z)} - a_i(T_{MID}^{(x)})) \\ \text{sign}(O_j^{(z)} - a_{i-1}(O_j^{(x)})) = \text{sign}(T_{MID}^{(z)} - a_{i-1}(T_{MID}^{(x)})) \end{cases} \quad (15)$$

where j is the index of obstacle point.

For the remaining points, the distance between the section arc and obstacle points defined as follows:

$$R_j = |O_j C| - S/\theta \quad (16)$$

Collision occurred if the condition is met:

$$\min(|R_j|) < \lambda \quad (17)$$

where λ is a value that define width of the safe zone.

If a collision occurs then bending angle of the section must be increased in the case when $R_j > 0$ and decreased in the case when $R_j < 0$.

V. SIMULATION

The algorithm was tested on three-section continuum robot that moves along a trajectory inside the tube. Each section length 15 mm and has one degree of freedom — bending. The whole robot also has an additional degree of freedom: linear movement along Z axis. When collision occurs bending angle changed at 3° . Trajectory defined by 13 points. Other simulation conditions presented in Table I.

TABLE I. SIMULATION CONDITIONS

Linear tolerance, mm	ε	λ , mm	Maximum bending angle, °
0.1	$\theta/50$	0.6	90

VI. RESULTS

In this section, we present the simulation results. Simulations are carried out using MATLAB 2019a on a Windows 7 with Intel Core i7-4790K 4.00 GHz CPU and 16.0 GB RAM.

Result of movement along trajectory inside the tube presented at Figure 4. Continuum robot successfully moves through tube without collisions. Operating time for whole trajectory was 0.27 s. Experiments show that λ lower than the

minimum distance between obstacles points leads to false positive solutions due to robot passing between obstacles.

VII. CONCLUSION

In this paper, we proposed a method for collision avoidance for planar multi-section continuum robot. The method is able to determine the distance from the robot to the obstacle to define the collision occurred. The method also allows determining which movement needs to be performed by the continuum robot to avoid collisions. The calculation time of 13 points of the trajectory was 0.27 s. The operation suggests that the robot is capable of working in real time to avoid obstacles in a dynamically changing environment.

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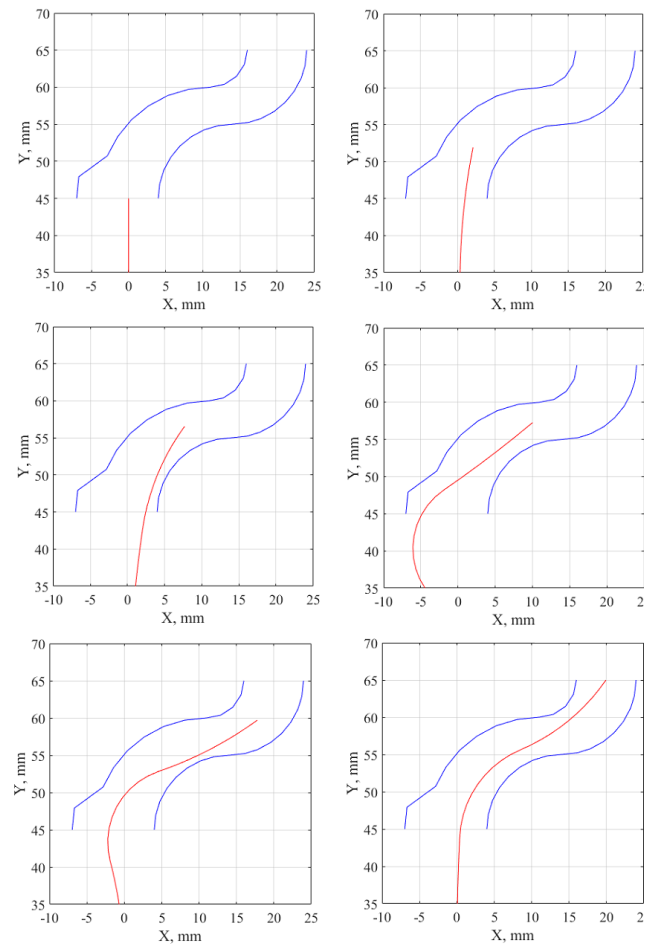


Fig. 4. Continuum follows a trajectory through a tube avoiding contact with walls.